

Esp32Tap Breadboard — Logical Cluster Layout

Approximate placement only. Use two full-size breadboards side-by-side; keep each zone compact, then connect zones by the named nets below. DigiKey order numbers and capacitor types are on page 3.

Start unpowered. Leave STANDALONE POWER and COIL POWER open. Meter-identify relay, transistor, regulator, and breakout pins before inserting them.

Breadboard A — treadmill / power side

+8V_RAW: short rail, connector end only

1 — RJ45 + relay contacts

- CONSOLE and MOTOR breakouts
- K1 at right edge of zone
- pins 1/7 → GND
- pins 2/8 → +8V_RAW
- pins 3/4/5 pass through
- pin 6 through K1 contacts

Put RJ45 breakouts at the outer edge.

+8V_RAW

2 — Input power

- RXEF075 + 1N5822
- P6KE12A TVS
- TSR1-2433E
- 22 μ F + two 10 μ F caps
- local VIN and TSR_3V3 strips

Keep protection parts near +8V entry.

VIN

4 — Voltage monitor

- TPS3700 breakout
- UV divider: 150k / 10k
- OV divider: 255k / 10k
- two 1 nF filters
- TREAD_OK pulls + GPIO6 R

Keep divider midpoints short.

K1 coil

5 — Relay driver

- TPS709 + 1 μ F / 4.7 μ F
- BC337 + 560 Ω + 10k
- P6KE6.8CA across K1 coil
- COIL POWER removable jumper

Place beside K1; verify B/C/E leads.

GND: join both boards with two jumpers

Breadboard B — ESP32 / logic side

LOGIC_3V3: logic board only

3 — ESP32 DevKit

Straddle center ditch; USB faces edge.

Left-facing jumpers:

- GPIO6 — TREAD_OK
- GPIO21 — RELAY_CMD
- GPIO15 — TX_ENABLE
- GPIO17 — UART TX
- GPIO18 / 16 — RX taps

Right-facing jumpers:

- GPIO4 / 5 — K1 feedback
- GPIO7 — VBUS_PRESENT_N
- GPIO38 — status LED

Rails:

- 3V3 → LOGIC_3V3
- GND → common GND
- 5V/VBUS → zone 7 only

Leave room for USB cable and reset.

GPIOs

6 — Logic + UART

- SN74AHC08N across ditch
- SN74AHC126N across ditch
- 100 nF at each IC
- command pull-downs
- R_TX 100 Ω
- two 10k RX resistors

IC notches point the same way.

Tie every unused input low.

7 — Feedback + LEDs

- K1 feedback pull-ups
- 2N7000 VBUS sense
- green status LED + 330 Ω
- red power LED + 1k
- C_LOGIC 100 nF

GPIO4/5/7/38

INTER-BOARD JUMPERS: SEE PAGE 2

Cluster Build Order and Inter-Zone Jumpers

Build and meter-check one zone at a time. Use the point-to-point wiring PDF for every pin-level connection.

Suggested build order

Order	Zone	Stop-and-check point
1	3 — ESP32	USB boot works; 3V3 and GND rails are correct.
2	7 — Feedback/LEDs	Power LED works; GPIO pins are not shorted.
3	6 — Logic/UART	DIP orientation, VCC/GND, unused inputs, and decouplers checked.
4	1 — RJ45/K1	Pass-through continuity correct with K1 unpowered.
5	2 — Input power	Polarity checked; VIN and GND have no short. Do not join standalone power yet.
6	4 — Monitor	Divider values and TREAD_OK pull resistors meter correctly.
7	5 — Relay driver	BC337 lead order and coil TVS checked; leave COIL POWER open.

Rail policy

Rail/net	Where	Rule
GND	Both boards	Join boards with two short jumpers; all USB and treadmill grounds are common.
+8V_RAW	Zone 1 → zone 2 only	Do not run along the logic board.
VIN	Zones 2, 4, 5	Keep on treadmill/power board.
LOGIC_3V3	Zones 3, 4, 6, 7	Powered from DevKit USB during initial bring-up.
TSR_3V3	Zone 2	Join to LOGIC_3V3 only through removable STANDALONE POWER jumper.
+5V_RLY	Zone 5	Short local strip only; coil connection stays removable.

Major inter-zone jumpers

Zones	Named nets / purpose
1 → 2	+8V_RAW, GND — protected local-power input.
2 → 4	VIN, GND — monitor supply and divider source.
2 ≠ 3	STANDALONE POWER — TSR_3V3 to LOGIC_3V3; install only when USB is absent.
3 ≠ 4	LOGIC_3V3, TREAD_OK_MCU/GPIO6.
4 → 6	TREAD_OK — hardware permission to both AHC08 gates.
3 → 6	GPIO21 RELAY_CMD, GPIO15 TX_ENABLE, GPIO17 UART TX.
6 → 5	RELAY_GATE — gated command to TPS709 enable and BC337 base resistor.
5 ≠ 1	K1 coil pins 1 and 16 — keep short; COIL POWER remains removable.
6 → 1	R_TX output to K1 pin 8 — switched ESP transmit.
1 → 3	CONSOLE pin 6 via 10k → GPIO18; pin-3 net via 10k → GPIO16.
1 → 7	K1 pins 9, 11, 13 — NO/COM/NC feedback contacts.
7 ≠ 3	GPIO4, GPIO5, GPIO7, GPIO38, 5V/VBUS, LOGIC_3V3.

Fast physical rules

- Put every DIP IC across a breadboard center ditch, with notches pointing the same direction.
- Put each 100 nF capacitor directly beside its IC supply pins.
- Keep UV_SENSE and OV_SENSE resistor/capacitor nodes inside zone 4.
- Keep BC337, its two resistors, coil TVS, and K1 coil pins together.
- Use colored jumpers consistently: black GND, red raw/VIN, blue 3V3, yellow control, green feedback.
- Label both ends of every jumper that crosses between boards.

This page groups the circuit; it does not replace the 126-row From/To wiring list. Never infer a physical pin from package shape—meter or datasheet-verify it.

DigiKey Order Cross-Reference

Matched to DigiKey invoice **129832848**, sales order **100614250**, dated 27-Jul-2026. **NOT ORDERED** means not present on this invoice; it does not mean you do not already own it.

Ordered semiconductors, modules, and hardware

Zone / use	Manufacturer part	DigiKey part number
3 — DevKit	ESP32-S3-DEVKITC-1-N8R8	1965-ESP32-S3-DEVKITC-1-N8R8-ND
4 — monitor	TPS3700DDCR	296-30395-1-ND
4/5 — SOT-23 adapters (2)	LCQT-SOT23-6	A879AR-ND
5 — relay regulator	TPS70950DBVR	296-35484-1-ND
6 — AND gates	SN74AHC08N	296-4524-5-ND
6 — tri-state buffer	SN74AHC126N	296-4535-5-ND
1 — relay K1	G5V-2 DC5	39-G5V-2DC5-ND
5 — relay transistor	BC337-40	4878-BC337-40CT-ND
7 — VBUS MOSFET	2N7000	4878-2N7000CT-ND
2 — Schottky diode	1N5822-TP	1N5822-TPMSCT-ND
5 — coil TVS	P6KE6.8CA	18-P6KE6.8CACT-ND
2 — resettable fuse	RXEF075	RXEF075HF-ND
2 — 3.3 V converter	TSR 1-2433E	1951-TSR1-2433E-ND
2 — input TVS	P6KE12A-TP	P6KE12A-TPMSCT-ND
7 — green status LED	LTL-4233	160-1130-ND
7 — red power LED	151051RS11000	732-5016-ND
Adapter/header use	PRPC040SAAN-RC, 40-position	S1011EC-40-ND

Ordered resistor values

Use	Value / type	DigiKey part number
BC337 base	560 Ω, 1%, 1/4 W axial	MFR-25FBF52-560R-ND
UV divider top	150 kΩ, 1%, 1/4 W axial	150KXBK-ND
OV divider top	255 kΩ, 1%, 1/4 W axial	13-MFR-25FBF52-255K-ND

Capacitors from the order

Circuit use	Value and capacitor type	DigiKey part number
Logic/IC bypasses (4)	100 nF, 50 V, X7R ceramic radial, non-polar	BC1084CT-ND
TPS709 input (1)	1 μF, 25 V, X7R ceramic radial, non-polar	445-173583-1-ND
UV/OV filters (2)	1 nF, 50 V, C0G ceramic radial, non-polar	445-173473-1-ND
VIN bulk (1)	22 μF, 25 V, aluminum electrolytic radial, polarized	732-8821-1-ND
TSR input/output (2)	10 μF, 50 V, aluminum electrolytic radial, polarized	1189-2322-ND
TPS709 output (1)	4.7 μF, 50 V, aluminum electrolytic radial, polarized	732-8660-1-ND

Polarity: electrolytic long lead is positive; the body stripe marks negative. The ceramic capacitors above are non-polar.

Required but not on this DigiKey invoice

Required item	Approx. quantity / use	Order status
10 kΩ resistors	12 — dividers, pulls, RX taps, feedback, VBUS	NOT ORDERED
100 kΩ resistor	1 — TREAD_OK pull-down	NOT ORDERED
4.7 kΩ resistor	1 — GPIO6 isolation	NOT ORDERED
100 Ω resistor	1 — UART TX series resistor	NOT ORDERED
330 Ω resistor	1 — green status LED	NOT ORDERED
1 kΩ resistor	1 — red power LED	NOT ORDERED
RJ45 breakout boards	2 — CONSOLE and MOTOR	NOT ORDERED
Full-size breadboards	2 recommended for this layout	NOT ORDERED
Jumper-wire assortment	126 point-to-point connections	NOT ORDERED
Removable jumper/shunts	2 — STANDALONE POWER and COIL POWER	NOT ORDERED

Do not substitute capacitor type or voltage rating casually. Preserve the listed ceramic dielectric and electrolytic polarity; use equal or higher voltage ratings.